

Thinking-Aware Baselines and Low-Data Post-Training for Financial Numerical Reasoning

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Abstract

Financial numerical reasoning requires models to integrate evidence from hybrid text-table inputs and perform multi-step arithmetic. Using a locked evaluation protocol on FinQA, we compare Qwen3-4B and Qwen3-8B under zero-shot and low-data task-specific supervised fine-tuning (SFT). Contrary to the expectation that larger models are more adaptable, we find that SFT consistently improves Qwen3-4B (zero-shot 24.93% $\rightarrow$ best SFT 32.43%) but fails to recover the weaker zero-shot performance of Qwen3-8B (20.40% $\rightarrow$ best SFT 18.74%). The 8B model remains below its own zero-shot baseline across all data sizes and training lengths. Error analysis shows that 4B gains are driven by improved format compliance, while 8B failures stem from persistent numerical reasoning errors, not parsing issues. Our results highlight that prompt-format alignment and model-specific conversational priors are decisive factors in low-resource SFT for numerical reasoning, and that scale alone does not guarantee effective task adaptation.

1 Introduction

Financial question answering requires more than retrieving a text span or matching a table pattern. In FinQA, models must identify relevant evidence from hybrid text-table inputs and perform multi-step numerical reasoning to produce the correct answer. Because correctness is sensitive to scale, percentages, ratios, signs, and magnitude selection, FinQA provides a useful benchmark for testing whether language models can reason over financial data rather than rely on surface patterns (Chen et al., 2021; Zhu et al., 2021).

Despite rapid progress in large language models, financial numerical reasoning remains difficult. Prior work on FinQA and related benchmarks shows that performance depends on both

evidence grounding across heterogeneous inputs and arithmetic composition over tables and text (Chen et al., 2021; Zhu et al., 2021; Zhao et al., 2022; Chen et al., 2022; Reddy et al., 2024). Under a fixed evaluation protocol—no-thinking inference, a controlled final-answer format, and mathematical answer verification—we observe an unexpected result: Qwen3-4B outperforms Qwen3-8B in the zero-shot setting. This raises a concrete question: does the weaker baseline of the larger model reflect a fundamental limitation, or weaker task alignment recoverable through lightweight supervision?

We study this question through low-data task-specific supervised fine-tuning (SFT). Rather than assuming that 8B should eventually surpass 4B, we ask whether it is more recoverable under controlled adaptation. Prior work suggests that downstream performance depends not only on model scale, but also on adaptation design, supervision format, and data quality (Wei et al., 2022; Chung et al., 2024; Longpre et al., 2023; Dettmers et al., 2023). If 8B improves substantially more than 4B after SFT, its weaker zero-shot performance is more likely due to under-alignment than lack of usable task capacity. If both improve similarly, larger scale may offer no special advantage for adaptation. If 8B improves only marginally, lightweight alignment is unlikely to explain the original gap.

To test this, we conduct a protocol-locked comparison between Qwen3-4B and Qwen3-8B on FinQA. We fix the zero-shot anchor table under the same inference and evaluation setup, apply matched task-specific SFT to both models, and compare recovery from zero-shot baselines rather than post-training scores alone. We further run a data-size ablation to measure how much supervision is needed before meaningful gains

appear, and analyze error shifts to identify which failure types are repaired by SFT. Overall, we use FinQA as a controlled testbed for studying how model scale interacts with task-specific adaptation in financial numerical reasoning.

2 Related Work

2.1 Financial numerical reasoning and FinQA-style benchmarks

Financial question answering is a challenging numerical reasoning setting because answers must be grounded in both tables and text. FinQA is one of the earliest benchmarks designed for this problem, pairing questions over financial reports with gold reasoning programs and showing that the task remains difficult for strong pretrained models (Chen et al., 2021). TAT-QA extends this setting to hybrid tabular and textual financial content, with many questions requiring explicit arithmetic operations and multi-step composition (Zhu et al., 2021). Subsequent datasets expand the difficulty along different dimensions: MultiHiertt introduces multi-hierarchical tables and longer contexts (Zhao et al., 2022), ConvFinQA extends the task to conversational settings (Chen et al., 2022), and DocFinQA pushes it toward long-context document reasoning (Reddy et al., 2024). Together, these benchmarks establish financial numerical reasoning as a task requiring both evidence grounding and numerical composition.

2.2 Hybrid table-text numerical reasoning

Our work is also related to numerical reasoning over hybrid text-table inputs beyond finance. Prior work shows that standard reading-comprehension or table-question-answering methods are insufficient when answers require both evidence selection and arithmetic reasoning across modalities. TAT-QA makes this explicit in finance (Zhu et al., 2021), MultiHiertt shows further degradation when reasoning spans multiple hierarchical tables and paragraphs (Zhao et al., 2022), and RegHNT formulates the problem as expression generation over relations among numbers, schema, and supporting text (Lei et al., 2022). Unlike this line of work, we do not propose a new architecture for hybrid reasoning; we use FinQA as a controlled testbed for studying task-specific adaptation.

2.3 Supervised fine-tuning and instruction tuning for task alignment

Our study is also connected to instruction tuning and supervised fine-tuning as approaches to downstream task alignment. FLAN shows that finetuning on instruction-formatted task mixtures can substantially improve zero-shot generalization (Wei et al., 2022). Follow-up work shows that these gains depend on factors such as task diversity, prompt variety, balancing strategies, and chain-of-thought-style supervision (Chung et al., 2024; Longpre et al., 2023). At the method level, our setup follows the standard parameter-efficient finetuning paradigm: LoRA introduces low-rank adapters while freezing most pretrained weights (Hu et al., 2022), and QLoRA combines low-rank adaptation with quantized base models to enable efficient finetuning under limited memory budgets (Detrmers et al., 2023). These methods make matched 4B/8B comparisons feasible in our setting.

2.4 Model scale and task adaptation

A common assumption in large language model evaluation is that larger models should perform better on downstream reasoning tasks. Prior work on instruction tuning and adaptation suggests a more qualified picture. FLAN and subsequent work show that adaptation quality depends not only on scale, but also on task mixture, prompt design, and supervision format (Wei et al., 2022; Chung et al., 2024; Longpre et al., 2023). QLoRA further reports that dataset suitability and data quality can matter more than raw dataset size for downstream performance (Detrmers et al., 2023). More recently, selective fine-tuning work argues that standard SFT can improve task-specific performance while reducing generalization (Gupta et al., 2025). This literature motivates our central question: when a larger model underperforms a smaller one on a specialized benchmark such as FinQA, does the gap reflect a fixed capacity limitation or a task-alignment problem that targeted supervision can partially repair?

We study this question through the lens of *recoverability*: rather than assuming that the larger model should dominate after adaptation, we compare how much the 4B and 8B models recover from their zero-shot baselines under the same protocol, data, and finetuning design.

3 Method

We study whether low-data task-specific SFT can recover the zero-shot underperformance of Qwen3-8B relative to Qwen3-4B on FinQA. Our design measures recovery relative to each model’s zero-shot baseline and analyses which failure types are repaired by SFT.

3.1 Evaluation Protocol

We lock a single evaluation protocol for all runs so that zero-shot and SFT results are directly comparable. Inference uses no-thinking mode, which disables the model’s internal reasoning trace and forces a single-pass output. The model must place its answer inside a tag of the form `[FINAL_ANSWER]<number>[/FINAL_ANSWER]`, from which the numeric value is extracted and scored with math-verify. Math-verify performs symbolic numeric comparison with a small tolerance and handles percentage rescaling automatically.

We report three metrics. Strict accuracy (`acc_base`) counts a prediction as correct only under exact numeric match. Adjusted accuracy (`acc_adjusted`) additionally accepts predictions that match the gold answer after percentage rescaling (e.g. 0.14 and 14% are treated as equivalent). The parse failure rate (`parse_fail`) reports the fraction of predictions from which no numeric answer can be extracted, serving as a diagnostic for format compliance.

3.2 Parameter-Efficient Fine-Tuning

We fine-tune both models with parameter-efficient methods rather than full-model training: LoRA for Qwen3-4B, which inserts low-rank adapter matrices into the attention projections while keeping the base model frozen, and QLoRA for Qwen3-8B, which combines LoRA with a 4-bit quantised base model. The latter is required to fit 8B within our single-GPU memory budget. Hyperparameters are matched across both models wherever possible (Table 1); the two unmatched choices—adaptation method and learning rate—are treated as confounds and revisited in the Limitations section.

3.3 Supervision Formats

We compare two supervision formats to test whether a lightweight explicit reasoning target helps the model ground its numeric output. The

`answer_only` format supervises only the final tagged answer. The `formula_rationale` format adds a single inline formula line before the tagged answer, without free-form reasoning prose. This design keeps supervision short and fully grounded in the numeric values of the example, so any gain over answer-only can be attributed to the formula target rather than to extra unconstrained text.

3.4 Prompt Protocol Alignment

An early pipeline version used a plain-text template (`stage1_train_text`) with explicit Instruction/Input/Output fields at training time but a chat-style template (`chat_default`) at evaluation time. This mismatch alone was sufficient to eliminate SFT gains entirely. We therefore added a prompt protocol option to the evaluation script and evaluate each SFT checkpoint under the format that matches its training template. We treat prompt-protocol alignment as a methodological requirement rather than an implementation detail.

3.5 Analysis Procedures

Our analysis has three components. The *main comparison* measures how much each model’s accuracy changes from its zero-shot baseline after SFT and compares these deltas between 4B and 8B. The *data ablation* reports accuracy at nested training sizes (250, 1,000, full) under a short 100-step schedule, plus a training-length check that re-trains one full-data configuration for 1,640 steps to separate the effect of data size from training duration. The *error shift analysis* classifies failures into four categories—`parse_fail` (output cannot be parsed into a number), `numeric_mismatch` (extracted number is wrong), `percent_scaling` (correct value but wrong scale form, e.g. 0.14 vs. 14%), and `unit_confusion` (magnitude or unit error)—and compares their distributions before and after SFT.

4 Experiment

4.1 Dataset and Preprocessing

We use FinQA as the sole benchmark for both training and evaluation. Our starting training pool was a merged corpus of 25,185 examples drawn from FinQA, ConvFinQA, and MultiHiertt. An initial audit revealed two problems: the three source datasets use different answer conventions, introducing stylistic variation unrelated to nu-

merical reasoning, and around 32% of examples contained a mismatch between the annotated answer and the value obtained by executing the gold formula (`formula_exec_ok=false` or `scale_relation≠consistent`).

We built a stricter training subset (`finqa_clean`) by retaining only FinQA examples with `formula_exec_ok=true` and `scale_relation=consistent`. This yielded 3,277 training and 475 development examples. The cleaned subset preserves the original distribution of single-step, two-step, and multi-step reasoning questions. Evaluation uses the full FinQA test split of 1,147 examples.

4.2 Data-Size Ablation and Training-Length Check

We construct two smaller nested subsets of 250 and 1,000 examples from the cleaned pool (`train_250 ⊂ train_1000 ⊂ full`), stratified by the number of reasoning steps in the gold program (`seed=42`) to preserve the step-count distribution at each size. All ablation runs use a short 100-step schedule, corresponding to roughly 12% of one epoch, so that training-size effects are measured under a fixed low-resource budget. We additionally retrain the full-data 8B `answer_only` configuration for 1,640 steps (two full epochs) to separate the effect of adding more data from the effect of training longer on the same data.

4.3 Models and Training Configuration

We compare Qwen3-4B and Qwen3-8B, loaded from the public Qwen3 checkpoints on Hugging Face. All experiments were run on a single NVIDIA GB10 GPU with 121.7 GB of memory. Training uses the TRL SFTTrainer with PEFT and BF16 mixed precision; only the LoRA adapters on the attention projections (`q_proj`, `v_proj`) are trained while base model weights remain frozen.

Table 1 lists the full configuration. Shared settings include LoRA rank 16, alpha 32, dropout 0.05, batch size 4, max sequence length 512, and 2 training epochs. The two unmatched settings are the adaptation method (LoRA for 4B, QLoRA with 4-bit quantisation for 8B) and the learning rate ($5e-5$ for 4B, $2e-4$ for 8B, following the common practice of using a higher rate to compensate for quantisation).

4.4 Evaluation Setup

All models are evaluated on the FinQA test split ($n=1,147$) under the locked protocol described in Section 3.1. The primary context setting is oracle, which supplies only the gold supporting evidence for each question; we also report a full-context check as a secondary comparison. Inference uses no-thinking mode with a maximum of 256 new tokens. Scoring uses math-verify with absolute and relative tolerances of 0.001 and automatic percentage rescaling enabled. All SFT checkpoints are evaluated under the prompt protocol matching their training format.

5 Results

All results are on the FinQA test split ($n=1,147$) under the locked evaluation protocol described in Section 3.1.

5.1 Main Comparison

Under the locked protocol, Qwen3-4B and Qwen3-8B respond to task-specific SFT in opposite directions (Table 3; Figure 1). For Qwen3-4B, the zero-shot text-prompt baseline reaches `acc_base` = 24.93% with `parse_fail` = 8.98%. After SFT on the 3,277-example cleaned subset, `acc_base` rises to 31.21% under `answer_only` (+6.28pp) and 32.43% under `formula_rationale` (+7.50pp). The `formula_rationale` configuration gives the best overall result for 4B.

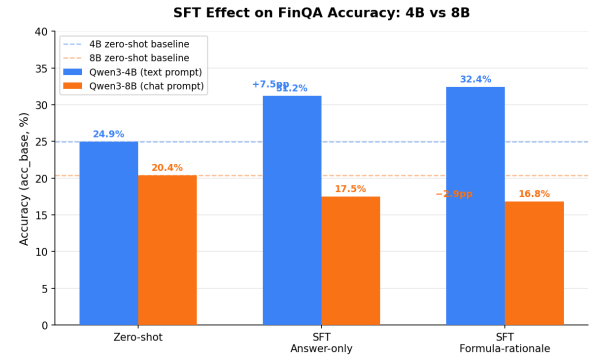


Figure 1: Effect of task-specific SFT on FinQA accuracy for Qwen3-4B and Qwen3-8B.

Qwen3-8B shows the opposite pattern. Its zero-shot chat-format baseline (Table 2) reaches `acc_adjusted` = 20.40% with `parse_fail` = 3.14%, and no SFT configuration surpasses it. The best 8B SFT result (full data, 1,640 steps, `answer_only`) reaches `acc_adjusted`

Setting	Qwen3-4B	Qwen3-8B
Adaptation method	LoRA	QLoRA (4-bit)
LoRA rank	16	16
LoRA alpha	32	32
LoRA dropout	0.05	0.05
Target modules	q_proj, v_proj	q_proj, v_proj
Learning rate	5e-5	2e-4
Epochs	2	2
Batch size	4	4
Max sequence length	512	512
Precision	BF16	BF16

Table 1: Training configuration for Qwen3-4B and Qwen3-8B. All settings are matched except for the adaptation method and the learning rate.

Model	Thinking	acc_base (%)	acc_adjusted (%)	parse_fail (%)	percent_recovered	tag (closed / open / absent)
4B	false	12.99	31.39	4.27	211	985 / 136 / 26
8B	false	12.03	20.40	3.14	96	1144 / 2 / 1

Table 2: Zero-shot baseline results on FinQA under the chat-format oracle setting ($n = 1147$). Both models are evaluated with thinking=false.

Model	Format	Zero-shot	Best SFT	Delta	Parse fail (ZS / SFT)
Qwen3-4B	text	24.93	32.43	+7.50	8.98 / 10.29
Qwen3-8B	chat	20.40	18.74	-1.66	3.14 / 3.14

Table 3: Main comparison between zero-shot and best SFT result for each model on FinQA ($n=1,147$)

= 18.74%, 1.66pp below zero-shot. The two full-data 100-step runs perform worse still, at 17.52% (answer_only, -2.88pp) and 16.83% (formula_rationale, -3.57pp). SFT under matched conditions raises 4B above its baseline but keeps 8B below its own.

We vary both training size and training length for 8B and report acc_adjusted (Figure 2).

5.2 Data Ablation and Training-Length Check

Under the 100-step schedule, 8B reaches 15.95% at 250 examples (-4.45pp vs. zero-shot), 15.69% at 1,000 (-4.71pp), and 17.52% at full data (-2.88pp). Extending the full-data run to 1,640 steps raises accuracy to 18.74% (-1.66pp) but still does not close the gap. Across all four conditions, accuracy stays within a narrow 15.69%–18.74% band, always below the 20.40% zero-shot baseline. Neither scaling supervision nor extending training length alone recovers 8B performance in our configuration.

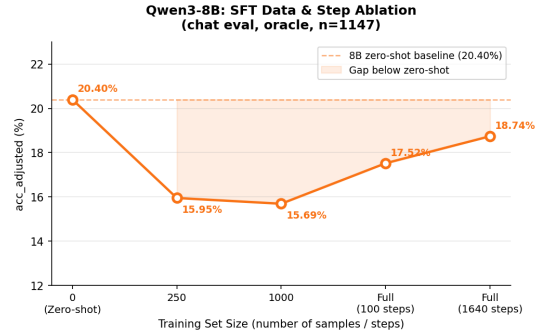


Figure 2: Data-size and training-step ablation for Qwen3-8B on FinQA. Dashed line indicates the zero-shot baseline (20.40%).

5.3 Error Shift Analysis

We examine how SFT changes the composition of errors rather than only headline accuracy (Figure 3; Table 4).

For Qwen3-4B, the most visible change is in format compliance: predictions with an absent FINAL_ANSWER tag drop from 409 under

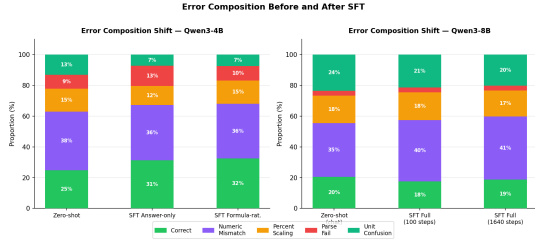


Figure 3: Error composition before and after SFT for Qwen3-4B (left) and Qwen3-8B (right).

zero-shot to 84–96 after SFT. The two supervision formats differ in their residual error profile: `formula_rationale` produces 3.6pp fewer `parse_fail` cases than `answer_only`, but 2.6pp more `percent_scaling` errors and 1.3pp more `unit_confusion` cases. For example, a question about pre-tax pension expense change yields gold -0.176 , but the 4B SFT output is `[FINAL_ANSWER] -17.6% [/FINAL_ANSWER]`—numerically correct but in the wrong scale form.

For Qwen3-8B, `parse_fail` stays flat at 3.14% across all conditions, confirming that the accuracy drop is not driven by format failures. SFT does reduce one output artifact—the pattern of echoing input context after the answer tag drops from 64% to 6%—but this does not translate into accuracy gains. The residual 8B failures are dominated by `numeric_mismatch` cases: for instance, a question asking for total impairment costs from 2003 to 2005 (gold = 5.8) receives the single-year 2005 value (1.3) instead of the three-year sum.

6 Discussion

The divergent responses of Qwen3-4B and Qwen3-8B to identical SFT reveal a stark contrast in *recoverability*. We argue that this contrast stems from an interaction between data quality, prompt alignment, and the model’s prior distribution, rather than a simple scale effect.

6.1 Why Does SFT Work for 4B but Not for 8B?

For Qwen3-4B, SFT succeeded only when two conditions were met: strict data cleaning and perfect alignment between training and evaluation prompts. This suggests 4B’s zero-shot weakness was primarily in format grounding, not reasoning

capacity. Under clean, aligned conditions, SFT effectively repaired the interface.

For Qwen3-8B, the situation reverses. Its zero-shot performance exhibits extreme prompt sensitivity, indicating a deeply embedded conversational prior. Plain-text SFT, by violating this prior, forces the model to resolve a format mismatch at the expense of reasoning fidelity. Although surface compliance improves (fewer echoes, consistent tagging), numerical accuracy does not recover. Low-data SFT appears insufficient to re-weight the internal pathways for arithmetic composition in a model with strong instruction-tuning priors.

6.2 Error Patterns as an Adaptation Diagnostic

Error-shift analysis supports this interpretation. For 4B, SFT primarily reduced parse failures, confirming that initial errors were largely syntactic. For 8B, parse failures were negligible even before SFT; the persistent deficit originates from numeric mismatches, especially on multi-step aggregations. This decoupling of format learning from reasoning improvement underscores that improving instruction-following does not automatically enhance numerical precision in larger models.

6.3 Methodological Takeaways

Two practical lessons emerge. First, prompt-protocol alignment is a first-class design variable. A mismatch can nullify SFT gains, particularly for chat-optimized models. Second, larger scale does not guarantee superior adaptability under low-data budgets. The *recoverability delta* may be a more informative selection criterion than absolute zero-shot scores.

6.4 Open Questions

Confounds remain. The use of QLoRA for 8B versus full LoRA for 4B cannot be entirely dismissed, though the observed format improvements suggest the adapter was functional. More critically, SFT within a native chat template remains untested; it is possible that format-aligned supervision would unlock 8B’s capacity. Whether these findings generalize beyond FinQA to other reasoning benchmarks is an open empirical question.

Model	Error type	Gold	Model output	Why it matters
Qwen3-4B	parse_fail	None	[FINAL_ANSWER] Yes [/FINAL_ANSWER] . . . [FINAL_ANSWER] No [/FINAL_ANSWER]	Multiple answer tags prevent the evaluator from parsing a single answer.
Qwen3-4B	percent_-scaling	-0.176	[FINAL_ANSWER] -17.6% [/FINAL_ANSWER]	Output is a percentage while gold expects a decimal; clearest scale-format trade-off after SFT.
Qwen3-8B	parse_fail	None	[FINAL_ANSWER] 435545 [/FINAL_ANSWER] + context echo	Tagged answer followed by repeated input context; format compliance is only partial.
Qwen3-8B	numeric_-mismatch	-35.0	[FINAL_ANSWER] -\$35 [/FINAL_ANSWER]	Math-verify interprets the output as positive 35 rather than -35.
Qwen3-8B	numeric_-mismatch	5.8	[FINAL_ANSWER] 1.3 [/FINAL_ANSWER]	Single-year value (1.3) instead of three-year sum (5.8); multi-step reasoning not restored.
Qwen3-8B	percent_-scaling	0.14	[FINAL_ANSWER] 14.5% [/FINAL_ANSWER]	Percentage instead of decimal; decreases under formula-rationale but offset by other failures.

Table 4: Representative error cases from the error-shift analysis.

7 Limitations

This study has several limitations that should be considered when interpreting the findings.

- Training steps and epoch coverage.** Most SFT runs (except the final 8B full-step condition) were limited to 100 steps, which corresponds to only 12% of one epoch over the 3,277 training samples. This was a deliberate choice to simulate low-resource SFT, but it means we cannot claim that 8B would fail under a longer, more conventional training schedule. The 1,640-step run (2 full epochs) did improve 8B’s accuracy (from 17.52% to 18.74%) but still did not surpass the zero-shot baseline. However, it is possible that even longer training or different learning rate schedules might close the gap.
- Quantization discrepancy between models.** Qwen3-4B was fine-tuned with full LoRA (no quantization), while Qwen3-8B used QLoRA (4-bit) due to GPU memory constraints. This difference could affect learning capacity and final performance. We cannot rule out that 8B with full LoRA might have achieved better results. Nevertheless, the fact that 8B SFT successfully reduced parse failures indicates that the QLoRA adapter did update the model; the failure is in reasoning, not in adaptation per se.
- Single training format (plain-text Instruction/Input/Output).** We did not experiment with SFT using the chat template (i.e., system + user messages) that matches Qwen3-8B’s preferred interface. The core finding—that plain-text SFT fails for 8B—does not imply that *no* SFT format can work. A follow-up using chat-formatted SFT is necessary to determine whether the failure is inherent to 8B or specific to the format mismatch.
- Limited supervision styles.** We compared only two target formats (answer-only and formula-rationale). More sophisticated supervision (e.g., chain-of-thought with explicit step-by-step calculations, or program-of-thought) might yield different results, especially for 8B.
- No cross-validation or statistical testing.** All results are reported on a single fixed test split (n=1147). We did not perform significance tests (e.g., bootstrap confidence intervals) due to the high cost of repeated evaluations, but the magnitude of differences (e.g., -1.66 pp for 8B full-step) is small relative to the zero-shot baseline. With a different random seed or test split, the difference might disappear or reverse.
- Single benchmark (FinQA).** FinQA is a numerical reasoning dataset over financial reports. The conclusions may not generalize to other reasoning tasks (e.g., arithmetic word problems, tabular QA) or other domains. The observed format sensitivity could be specific to the way FinQA questions are structured.

7. **No ablation of the “clean strict” filtering criteria.** Our clean subset required `source=finqa`, `formula_exec_ok=true`, and `scale_relation=consistent`. This removed 32% of the original FinQA training samples. While this ensured data quality, it also reduced the training set size. A relaxed filter might preserve more samples and potentially benefit 8B differently.
8. **Lack of chat-template SFT for 4B.** For completeness, we did not test whether 4B would also benefit from chat-formatted SFT. Since 4B already shows strong gains with plain-text SFT, this is not a critical gap, but it limits our ability to claim that 4B is inherently more robust.

Despite these limitations, the study provides clear, reproducible evidence that prompt-format alignment and model-specific priors are decisive factors in low-resource SFT for numerical reasoning. Future work should systematically compare SFT across multiple prompt templates and model families before concluding on a model’s “SFT-ability.”

8 Conclusion

This study shows that low resource task-specific SFT improves Qwen3-4B but fails to recover Qwen3-8B on FinQA. Under a controlled protocol, 4B achieves clear gains, where improvements mainly come from better format compliance and small changes in the error mix; while 8B remains below its zero-shot baseline across all settings, the failure is not driven by format issues but by persistent numerical reasoning errors, especially in multi-step computation. These results suggest that model scale alone does not guarantee effective adaptation. Instead, prompt format alignment and prior training behaviour play a key role. Future work should test format-aligned SFT, especially using chat-based training.

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A Prompt Templates

Table 5 shows the two prompt formats used in this study. The `stage1_train_text` format is a plain-text template used for both SFT training and prompt-aligned evaluation. The `chat_default` format is the chat-style template used for zero-shot evaluation of both models and for the chat-evaluated 8B SFT runs.

B Data Cleaning Details

Our data cleaning pipeline starts from a merged corpus of 25,185 examples from three sources: FinQA, ConvFinQA, and MultiHiertt. Table 6 summarises the filtering results. Table 7 shows the distribution of reasoning steps across training subsets, confirming that stratified sampling preserves the step-count distribution at each ablation size.

C Full Experimental Results

Table 8 reports accuracy and parse failure rates for all experimental conditions, including runs not detailed in the main text.

D Supervision Format Examples

Below are examples of the two supervision formats applied to the same training instance.

Answer-only format.

```
[FINAL_ANSWER] 3.8 [/FINAL_-
ANSWER]
```

Formula-rationale format.

```
Formula: (3.8 / (100 / 100))
[FINAL_ANSWER] 3.8 [/FINAL_-
ANSWER]
```

The formula-rationale format adds a single line showing the arithmetic expression derived from the gold program, without any free-form reasoning prose.

E Compute and Reproducibility

Hardware. All experiments were run on a single NVIDIA GB10 GPU with 121.7 GB of memory. Training the 4B model under LoRA used approximately 30 GB of GPU memory; the 8B model under QLoRA used approximately 55 GB.

Training time. A full 2-epoch training run on the 3,277-example cleaned subset takes roughly 1.5 hours for Qwen3-4B (LoRA) and 4 hours for Qwen3-8B (QLoRA). The extended 1,640-step run on 8B takes approximately 7 hours. Evaluation on the 1,147-example test split takes 20–40 minutes per run depending on model size.

Software. Training is implemented with the `TRL SFTTrainer` on top of `PEFT 0.11`, using `Hugging Face transformers`. Evaluation uses the `math-verify` library for symbolic numeric comparison. All code, configuration files, and experiment scripts are released with the paper.

Randomness. All training and data-subset sampling use a fixed random seed (`seed=42`). Reported results are from a single run per configuration; we did not perform multi-seed repetitions due to compute constraints (see Limitations).

Model and data licenses. Qwen3-4B and Qwen3-8B are released under the Apache-2.0 license. FinQA (Chen et al., 2021) is released for research use under the MIT license. Our use of these resources complies with their respective terms.

Protocol	Template
stage1_ train_text	Instruction: Solve the financial numerical reasoning problem. Return exactly one tagged final answer as [FINAL_ANSWER] . . . [/FINAL_ANSWER] in the output. Input : Context: <i>{context}</i> Question: <i>{question}</i> Output : <i>{answer or formula + answer}</i>
chat_ default	System: You are a helpful assistant. User: <i>{same instruction + input as above}</i>

Table 5: Prompt templates used for training and evaluation. The `Output` field is present only during training; at inference time, the model generates freely after the input.

Stage	Count
Starting pool (merged)	25,185
Dropped: non-FinQA source	18,934
Dropped: scale relation mismatch	2,778
Dropped: formula execution failure	196
Retained (train)	3,277
Retained (dev)	475

Table 6: Data cleaning statistics. The strict filter retains only FinQA examples with `formula_exec_ok=true` and `scale_relation=consistent`.

Subset	Single	Double	Multi
Full (3,277)	1,962 (59.9%)	873 (26.6%)	442 (13.5%)
1,000	599 (59.9%)	266 (26.6%)	135 (13.5%)
250	150 (60.0%)	66 (26.4%)	34 (13.6%)
Dev (475)	281 (59.2%)	139 (29.3%)	55 (11.6%)

Table 7: Distribution of reasoning steps (single, double, multi) across training subsets. Percentages confirm that stratified sampling preserves the step-count distribution.

Model	Condition	Prompt	Steps	acc_base (%)	acc_adj (%)	parse_fail (%)
<i>Zero-shot baselines</i>						
4B	zero-shot	chat	–	12.99	31.39	4.27
4B	zero-shot	text	–	24.93	32.78	8.98
8B	zero-shot	chat	–	12.03	20.40	3.14
8B	zero-shot	text	–	7.15	7.32	64.43
<i>SFT results (prompt-aligned evaluation)</i>						
4B	answer_only, full	text	100	31.21	33.22	13.08
4B	formula_rationale, full	text	100	32.43	33.83	10.29
8B	answer_only, 250	text	100	11.77	15.95	7.32
8B	answer_only, 1000	text	100	12.21	15.69	9.15
8B	answer_only, full	text	100	13.78	17.52	7.50
8B	formula_rationale, full	text	100	13.34	16.83	8.11
<i>SFT results (chat-format evaluation)</i>						
8B	answer_only, 250	chat	100	10.55	15.95	3.14
8B	answer_only, 1000	chat	100	10.64	15.69	3.14
8B	answer_only, full	chat	100	10.99	17.52	3.14
8B	formula_rationale, full	chat	100	11.07	16.83	3.14
<i>Extended training (1,640 steps)</i>						
8B	answer_only, full	text	1640	12.12	14.39	3.40
8B	answer_only, full	chat	1640	12.73	18.74	3.14

Table 8: Full experimental results across all conditions. “acc_adj” denotes adjusted accuracy after percentage rescaling. “Steps” indicates the number of training steps; “–” for zero-shot. Main-text tables report the oracle setting; all SFT rows here also use oracle context.